

Anna University

REGULAR EXAMINATION ? 2025

Engineering Mathematics-I ? Regular Examination, 2025

Subject Code: MA101 Semester: 1 Regulation: R23

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Matrices and Linear Algebra)

1. Find the rank and eigenvalues of the given matrix. Verify the Cayley-Hamilton theorem for the given matrix. [12 Marks]
2. Explain Matrices and Linear Algebra with a neat diagram/sketch and a suitable example. [12 Marks]
3. Describe the working principle of Matrices and Linear Algebra with necessary equations. [12 Marks]

SECTION B (Unit 2: Differential Calculus)

4. State Rolle's theorem and Lagrange's mean value theorem. Verify them for the given function. [12 Marks]
5. Define Differential Calculus. Explain its significance in Engineering Mathematics-I with a suitable example. [12 Marks]
6. Explain Differential Calculus with a neat diagram/sketch and a suitable example. [12 Marks]

SECTION C (Unit 3: Integral Calculus)

7. Evaluate the given double integral by changing the order of integration. [12 Marks]
8. Solve a problem related to Integral Calculus. Show all the steps clearly. [12 Marks]
9. Write short notes on Integral Calculus. [8 Marks]

SECTION D (Unit 4: Vector Calculus)

10. Using Green's theorem, evaluate the given line integral around the closed curve. State Stokes' theorem and Gauss divergence theorem. [12 Marks]
11. Compare and contrast the key concepts involved in Vector Calculus. [12 Marks]
12. Write short notes on Vector Calculus. [8 Marks]

SECTION E (Unit 5: Ordinary Differential Equations)

13. Solve the given second order linear differential equation with constant coefficients. [12 Marks]
14. Apply the fourth order Runge-Kutta method to solve the given initial value problem up to the required accuracy. [8 Marks]

15. Explain Ordinary Differential Equations with a neat diagram/sketch and a suitable example. [12 Marks]

*** End of Question Paper ***